



Contents lists available at ScienceDirect

Journal of Management Science and Engineering

journal homepage: www.keaipublishing.com/en/journals/journal-of-management-science-and-engineering/

Application of quantum computing in discrete portfolio optimization

Justus Shunza ^{a, b, *}, Mary Akinyemi ^{b, c, 1}, Chika Yinka-Banjo ^{a, b, 1}^a Operations Research & Information Engineering, Cornell University, 36 Hoy Road, Ithaca, 14853, New York, United States^b Department of Mathematics, University of Lagos, Akoka, Lagos, 101017, Lagos, Nigeria^c Department of Computer Sciences, University of Lagos, Akoka, Lagos, 101017, Lagos, Nigeria

ARTICLE INFO

Article history:

Received 7 May 2022

Received in revised form 16 November 2022

Accepted 12 February 2023

Available online 22 May 2023

Keywords:

Combinatorial optimization

Induced tadpole graph

Lattice

Qubits

Trajectory

ABSTRACT

This study proposes a novel and more efficient quantum algorithm for portfolio optimization using quantum combinatorial optimization (QCO) techniques. A recent construction developed in 2021 has sparked the field of financial portfolio optimization through the Quantum Walk Optimization Algorithm (QWOA). In this study, we investigated the complexity and efficiency of quantum optimization algorithms with a special interest in QWOA. The objective is to minimize investment risk by having a good combination of assets in the portfolio. We also focused on reducing the number of iterations while attaining a high-quality resolution through contraction of the solution space to ease computations. The concept of QWOA was extended by constructing a newly outperforming scheme known as the “Quantum Mix Optimization Algorithm (QMOA).” QMOA algorithm codes were provided for the implementation and simulation of numerical results. In addition, the efficiency of QMOA, which is better than the existing QCO algorithms, was discussed. For instance, the least QWOA number of computations required to execute the initial state equation was $p > 2$, whereas this value was $p \geq 2$ in the proposed QMOA.

© 2023 China Science Publishing & Media Ltd. Publishing Services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

In finance, a portfolio is a collection of investments. The core components of financial investments include bonds, stocks (company shares), commodities, and cash. To maintain balance, an investor requires a good combination of assets, which accelerates the growth of capital and reduce investment risks (Marzec, 2014; Renumadhavi et al., 2021). There are three major types of portfolios, namely, growth, income, and value portfolios. In this study, the general structure of quantum portfolio optimization is analyzed. The intuition is derived from the underlying principles of portfolio choice, structuring the choices of investors, and the problem of selecting the best portfolio. Portfolio selection is simpler for a static portfolio choice problem than a dynamic problem. This is because a static problem only contains two periods, whereas a dynamic problem involves multiple periods, thereby making it challenging to predict the value of an asset from one period to the next until the n^{th} period

* Corresponding author.

E-mail address: js2996@cornell.edu (J. Shunza).

¹ These authors contributed equally to this work.

in the sequence. An investor privy to the discounting factor of an asset can easily compute the projected value of the asset at the n^{th} period.

According to the historical behavior of financial advisers or investors on efficient portfolios, they are never content and are averse to taking big risks (Han and Kim, 2000). That is, investors prefer portfolios that yield higher returns and lower variance. In the selection, financial advisers focus on the set of best portfolios based on their ability to yield higher returns while incurring lower risks, and drops the rest of the portfolios in the pool. In this study, we focus more on dynamic portfolio optimization, in which the problem can either be deterministic or stochastic. In the deterministic case (perfect foresight), there is no randomness in the development of a model to predict the future states of the system. However, in the stochastic case (uncertainty), random variation in one or more input(s) exists. For this case, standard time-series techniques are applicable.

The dynamic optimization problem is one of the most challenging problems to solve (orus et al., 2019). Over decades, the exercise of portfolio selection has continually troubled financial advisers over which portfolio should be selected among others based on the return it yields and the potential risk to be incurred. Recently, portfolio optimization algorithms have been designed to aid in portfolio selection. However, the efficiency and accuracy of most of the constructed systems are not perfect. Currently, there are probably more assets included in the construction of each portfolio. This process makes the portfolio structure complex and also increases the size of the selection set, as there can be several portfolios constructed from N assets (see Equation (1)).

$$\text{Possiblenumberofportfolios} = 2^N \text{ for } N \text{ assets.} \quad (1)$$

The portfolio selection process is an NP-hard problem. Recently, quantum computing has become a better candidate for portfolio optimization. The theory of combinatorial optimization was first discovered by a mathematician, Jack Edmonds. For decades, this concept has been widely used in various areas of optimization and has flourished in finance. Importantly, the emergence of quantum combinatorial optimization techniques has even improved optimization because quantum algorithms are provably strong and can solve NP-hard problems in polynomial time (Han and Kim, 2000; Mugel et al., 2020). Although there are several quantum algorithms recently developed to solve portfolio optimization problems, the Quantum Walk Optimization Algorithm (QWOA) has certain fascinating similarities with our newly proposed scheme, QMOA.

In addition, the Quantum Approximate Optimization Algorithm (QAOA) and Quantum Alternating Operator Ansatz (QAOAz) by Hodson et al. (2019a) laid a solid foundation for hybrid quantum-classical algorithms in financial portfolio rebalancing problems. These developments have led to a new realization of the applications of quantum computing in portfolio optimization. The Quantum Approximate Optimization Algorithm (QAOA) presents an initial state equation defined by

$$|\psi_f\rangle = \prod_{i=1}^p U_{\text{mix}}(t_i) U_C(\gamma_i) |\psi_0\rangle. \quad (2)$$

For a prescription of parameters, refer to Slate et al. (2021). QAOAz provides a proper structure of the Hamiltonian operator C in Equation (2). However, the range of p , which is the number of QAOAz iterations, is not provided (Cook et al., 2020). Analysis of the numerical results shows that the QWOA algorithm outperforms both QAOA and QAOAz by considering $p > 2$. However, the supremum of p for QWOA is not mentioned. In addition, the lattice of QWOA is an open network resembling a tree diagram starting from a fixed point and opening outwards as the number of assets increases. This leads to a complex and large solution space, making problem optimization more challenging. This study aims to address these challenges robustly through the following objectives:

1. Determine the supremum of p for the QWOA algorithm.
2. Reduce the solution space by designing an enhanced network (QMOA lattice).
3. Review the computation complexity and effectiveness of algorithms (determine the efficiency of QMOA).

Our goal is to select a portfolio whose properties yield the lowest risk and a net return greater than zero. The rest of the sections are organized as follows: Section 2 details the methods used to achieve the objective. Theoretical results with rigorous proofs are provided. Section 3 exposes the theoretical performance of the new algorithm (QMOA) and various results that emerge from the algorithm. This section also presents the outcome or rather the expected return of each portfolio for which the results influence the decision of an investor. Moreover, in Section 4, the algorithm construction and implementation process are discussed, and a clear strategy for the model improvement is illustrated by providing QMOA algorithm codes. We also extend the discussion regarding the efficiency and computational ability of well-known algorithms in this section. Finally, Section 5 concludes and recommends the implementation process of QMOA. It also provides an open question and opportunity for future research.

2. Methodology

In this section, the concept of portfolio construction is briefly built from Markowitz's portfolio selection process as described in (Markowitz, 1952). We also construct a random network of assets and reconstruct a lattice from the initial and final states whose design is modified to fit the quantum algorithms operations.

2.1. Portfolio construction: generalization for N -assets portfolio

Generally, we define the portfolio for N assets as

$$\prod_N = \sum_{i=1}^N \lambda_i S_i \text{ such that } \sum_{i=1}^N \lambda_i = 1. \quad (3)$$

Note that $\prod_N$ in Equation (3) does not imply that all portfolios allocate a fraction of wealth to every asset. Certain portfolios may have less than N assets in the combination but a minimum of two assets to avoid short selling. If $\text{Card}\{\prod\}$ is the cardinality of assets in portfolio $\prod_N$, then $2 \leq \text{Card}\{\prod\} \leq N$. Without loss of generality, if asset S_i is excluded in the mix, then its wealth fraction $\lambda_i = 0$ except for the special case where $N = 2$.

The generalization of expected return is expressed in Equation (4).

$$\bar{R}_{\pi_N} = \mathbb{E}[R_{\pi_N}] = \sum_{i=1}^N \lambda_i \mathbb{E}[R_i] = \sum_{i=1}^N \lambda_i \bar{R}_i. \quad (4)$$

Subsequently, the variance is generalized as follows:

$$\sigma_{\pi}^2 = \sum_{i=1}^N \lambda_i^2 \sigma_i^2 + 2 \sum_{i=1}^N \sum_{j=1}^N \rho_{ij} (\lambda_i \sigma_i) (\lambda_j \sigma_j) \quad \forall i \neq j. \quad (5)$$

If the assets are moving independently, then all the correlation coefficients ρ_{ij} for $i \neq j$ reduce to zero (MilneMaxwell Rounds, 2017). Hence, we have a perfect square as shown in Equation (6).

$$\sigma_{\pi}^2 = \sum_{i=1}^N \lambda_i^2 \sigma_i^2. \quad (6)$$

Suppose that the wealth is uniformly distributed among the N risks, then $\lambda_i = \frac{1}{N}$. This implies that

$$\sigma_{\pi}^2 = \frac{1}{N} \left(\frac{1}{N} \sum_{i=1}^N \sigma_i^2 \right) = \frac{1}{N} \mathbb{E}[\sigma_i^2]. \quad (7)$$

It is evident that Equation (7) tends to zero as the value of N approaches infinity, which reveals that the variance or the risk associated with a portfolio whose assets are uncorrelated disappears in the framework as the number of assets increases. In practice, assets are always correlated; that is, correlations and covariances of most assets are positive, implying that the risk cannot be equal to zero. It is possible for the risk of the portfolio to be less than the variance of an individual asset; that is, $\rho_{ij} \neq 0$. Given a uniformly distributed set of assets, that is, $\lambda_i = \frac{1}{N}$, Equation (5) reduces to

$$\sigma_{\pi}^2 = \sum_{i=1}^N \frac{1}{N^2} \sigma_i^2 + 2 \sum_{i=1}^N \sum_{j=1}^N \frac{\rho_{ij}}{N^2} \sigma_i \sigma_j \quad \forall i \neq j. \quad (8)$$

By the definition of the mean value (Kim et al., 2015) and factoring N , Equation (8) becomes

$$\sigma_{\pi}^2 = \frac{1}{N} \mathbb{E}[\sigma_i^2] + \frac{N-1}{N} \times \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=1}^N \rho_{ij} \sigma_i \sigma_j \quad \forall i \neq j. \quad (9)$$

If each pair of assets are perfect positive correlated, then the correlation coefficient $\rho_{ij} = 1$ for all $i \neq j$. The normalization of $\rho_{ij} \sigma_i \sigma_j$ is achieved by setting $\sum_i \sum_{j < i} \sigma_i \sigma_j = N(N-1)$ and scaling it by $\frac{1}{2N(N-1)}$. This is defined by doubling the sum $N(N-1)$ (Rebentrost et al., 2018) in Equation (9) with $\mathbb{E}[\sigma_i^2] = 1$. In this setting, the investor should be concerned about assets that are widely unrelated to minimize portfolio-wide risk. A portfolio whose assets holding are positively correlated tends to increase

the risk of loss. Therefore, financial advisers should be more careful when handling asset variability that leads to the convergence of Equations (9) and (10).

$$\sigma_{\pi}^2 = \underbrace{\frac{1}{N}}_{\text{mean variance}} + \underbrace{\frac{N-1}{N}}_{\text{mean covariance}}. \quad (10)$$

Unlike in Equation (7), it is clear that as the number of assets tends to infinity, the variance of the portfolio corresponds to the average covariance. We can then vary the individual risk of assets while maintaining covariance risk.

2.2. Dynamic discrete portfolio optimization

In this section, we present a random distribution of N assets geometrically and establish the linear relationship (covariance) between individual pairs of assets in the mix. The aim is to transform this problem into an optimization problem resembling the “Euler’s Tour” (Hurkens and Woeginger, 2004; Lawler, 1985) and deploy quantum combinatorial optimization techniques for optimization. Consider the lattice of weighted edges (covariances) of random asset mix in Fig. 1.

2.2.1. Formulation assumptions

1. The initial and final states are known in advance. First, the investor selects two preferred assets with distinct properties.
2. Assets in the portfolio follow a particular sequence or path based on the correlation coefficient ρ_{ij} between any two distinct pairs of assets i and j in the mix.
3. The value of each asset is well-known at the start of the period and remains constant within the period.
4. The investor is described as a risk-averse investor, as they only invest if the expected return is higher than the average return of the portfolio.

Fig. 1 depicts a lattice where every edge is assigned a weight (correlation coefficient) ρ_{ij} between the two assets S_i and S_j for $i, j \in [1, N]$. If $i = j$, then the returns R_i and R_j are said to be perfectly correlated. The edge $\mathcal{O}_{\pi}(i, i)$ is called the circuit or simply a self-loop, and $\text{Cov}(R_i, R_j) = \mathcal{O}_{\pi}(i, i) = 1$ for $i = j$. The operator $\mathcal{O}_{\pi}$ is an unbiased mapping that transforms the $\text{Cov}(R_i, R_j)$ into a correlation coefficient ρ_{ij} and is well-defined statistically. This map is unique because it estimates the covariance of any two nonadjacent assets or a series of nonadjacent assets. An instance of this is the covariance in red color in Fig. 1. From this figure, we rearrange the positions of assets and partition the lattice as follows in Fig. 2.

Fig. 1 represents a random network of N assets. We assume that the investor is interested to invest in $n_a < N$ assets. The task is to investigate a good mix of assets in the portfolio that guarantees the interest of an investor and minimizes risk or maximizes returns. This choice problem is complex, particularly for a large N . The intuition is to construct a better network of assets based on their correlations and a minimum of two types of assets preferred by the investor. The threshold of two assets ensures that the network has an initial and final state, a tadpole (Basavanagoud et al., 2017), if over two assets are preferred. This is the concept behind restructuring the network in Fig. 1 to form Fig. 2.

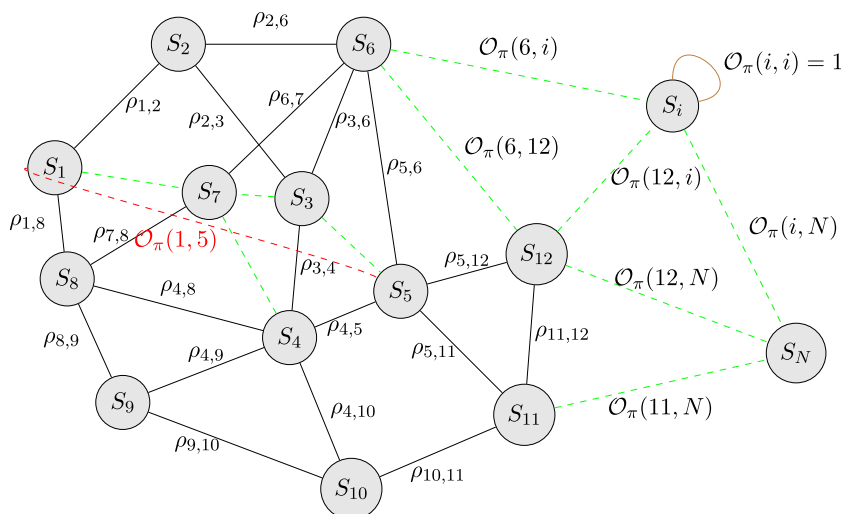


Fig. 1. Random distribution of N assets for $i \leq N$.

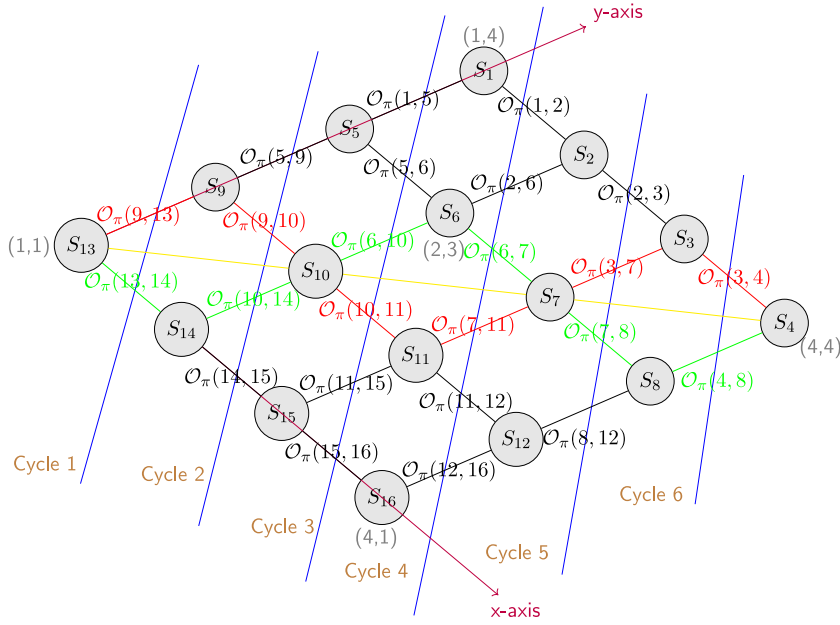


Fig. 2. Asset qubits set position.

3. Analytical results

In this section, we present our results based on the findings in Section 2. The optimal principle which proposes an effective approach that minimizes the number of calculations is discussed. Finally, we generalize the concept and convergence for an extremely large N .

3.1. Optimal computation

Definition 1. Principle of optimality

The principle of optimality states that if the control action $u(i), u(i+1), \dots, u(j), \dots, u(k)$ yields the optimal solution $x(i), x(i+1), \dots, x(j), \dots, x(k)$, then the truncated control $u(j), \dots, u(k)$ gives the optimal control solution for $x(j), \dots, x(k)$, respectively.

Therefore, it is unnecessary to solve the entire optimal control simultaneously. However, individual pieces can be solved at each cycle moving in the backward direction from the final state to the initial state by patching the pieces together. For instance, the possible paths from state S_{13} to state S_4 in Fig. 2 are given by Equation (11).

$$\binom{6}{3} = \left(\frac{6 \text{ movements}}{3 \text{ to upwards/downwards}} \right) = \frac{6!}{3!3!} = 20. \quad (11)$$

This reduces the number of iterations to only 20. To evaluate the effectiveness of this approach (Equation (11)), we estimate the number of possible iterations when solving this problem naturally. This is done by first computing the total correlation coefficient for all the possible paths that link S_{13} to S_4 and subsequently finding the least of all from this collection. From Fig. 2, the total number of edges in each cycle is Cycle 1 = 2, Cycle 2 = 4, Cycle 3 = 6, Cycle 4 = 6, Cycle 5 = 4, and Cycle 6 = 2. Therefore, there are $2 \times 4 \times 6 \times 6 \times 4 \times 2 = 2304$ calculations.

Definition 2. Let G be a geometrical construction of N assets in a network with nodes represented by the assets of a portfolio Π_N and edges corresponding to the covariances between pairs of assets. Furthermore, let $\varnothing_{old}$ be the number of calculations computed naturally from a collection of sums of correlation coefficients and $\varnothing_{m\text{-effective}}$ denote the number of iterations obtained by partitioning the lattice into m cycles. If the efficiency of a system is denoted by $\lambda_m \in (0, 1)$, we define a linear relationship between $\varnothing_{old}$ and $\varnothing_{m\text{-effective}}$ using Equation (12).

$$\varnothing_{m\text{-effective}} = \lambda_m \varnothing_{old}. \quad (12)$$

In general, let n be the minimum steps made from the initial state to the final state and k denote the steps in the same direction in a network. We compute the net calculations as follows:

$$\binom{n}{k} = \left(\frac{n \text{ total movements}}{k \text{ to same direction}} \right) = \frac{n!}{(n-k)!k!}. \quad (13)$$

Equation (13) possesses some interesting properties. First, the problem is fully a boolean choice at any point in the geometric setting. Second, we anticipate the possibility of having a special case of a network of assets, known as a “tadpole graph” described in (Weisstein, 2007). A tadpole graph comprises a connected cycle graph of at least three nodes with an extension of a path graph attached to the cycle graph by an edge (Basavanagoud et al., 2017). Without loss of generality, we define an induced tadpole in our lattice construction, which knocks out the general definition of a cycle graph and enables any network attached to the path graph. Thus, we only focus on the attachment of the path graph to the main network construction that represents the set of assets (excluding the initial and final states) selected by the investor at the beginning.

Definition 3. Let G^* be a network construction of N nodes (assets) and T_ω be its path graph. G^* is said to be a perfect network if there exists a number n such that $n^2 = N - \omega$, where ω is the number of nodes along the path graph.

Perfect networks are easy to construct and have good properties, which are enumerated below.

- The degree of all nodes is 3 except the initial and final states (see Fig. 1).
- The total number of nodes for adjacent cycles differs by 1.
- The total number of edges traversing any cycle is always even.

Corollary 4. Let G^* be a perfect network and C_N be the number of partitions of G^* into cycles, then C_N is defined by

$$C_N = 2(n-1) \quad \text{where} \quad n^2 = N - \omega. \quad (14)$$

Proof. The proof follows from the properties of a perfect network. The graph opens outwards from the initial state until it reaches the n^{th} cycle when it converges again to the final state. Because the difference between the total number of nodes in the adjacent cycles is 1, it follows that

$$N - \omega = \sum_{k=1}^n k + \sum_{k=1}^{n-1} k = n + 2 \sum_{k=1}^{n-1} k = n^2. \quad (15)$$

Note that $\sum_{k=1}^n k$ provides the total number of nodes in the network from the initial state to the n^{th} cycle, whereas $\sum_{k=1}^{n-1} k$ is the total number of nodes from the $(n+1)^{\text{th}}$ cycle to the final state. There are $n-2$ cycles in the latter because we do not consider the edges converging to the final state as a cycle. Thus, $C_N = 2(n-1)$.

Theorem 5. From Corollary 4, $N = 2n - 2 + \omega$. In addition, from Equation 13, the number of possible iterations of the system is

$$I_N^* = \left(\frac{2(n-1) + \omega}{(n-1) + 0.5\omega} \right) = \frac{(2n-2+\omega)!}{((n-1+0.5\omega)!)^2}. \quad (16)$$

Lemma 6. Let k_c be the number of edges traversing the c^{th} cycle for $c \in [1, \dots, C_N]$ and T_{tt} denote the natural number of calculations. Subsequently, T_{tt} is defined as the product of the sum of edges traversing every cycle in Equation 17.

$$T_{tt} = k_1 \cdot k_2 \cdot \dots \cdot k_{C_N}. \quad (17)$$

Conjecture 7. Given T_{tt} , as defined in Lemma 6 above, Inequality 18 holds.

$$T_{tt} > I_N^*, \text{ alternatively, } k_1 \cdot k_2 \cdot \dots \cdot k_{C_N} > I_N^*. \quad (18)$$

Definition 8. It is considered to be the best system if $\lambda_m \in (0, 1)$ is significantly small. The efficiency of the system defined in Equation (12) is estimated by computing λ_m as manifested in Equation (19).

$$\lambda_m \approx I_N^* \times T_{tt}^{-1}. \quad (19)$$

4. Algorithms and implementations

In this section, we briefly describe QAOA, QAOAz, and QWOA models. We also illustrate how the new scheme (QMOA) works and provide its implementation codes. To culminate the lattice into the quantum setting, we examine the formulation of the quantum Markowitz model and its properties, described in [Slate et al. \(2021\)](#), where $z = z_1 \cdot z_2 \cdots z_N$ is used to encode a particular choice of portfolio assets.

$$\text{Minimize: } c(z) = \lambda \sum_{i,j=1}^N \sigma_{ij} z_i z_j - (1 - \lambda) \sum_{i=1}^N \bar{R}_i z_i, \text{ subject to } \sum_{i=1}^N z_i = A. \quad (20)$$

4.1. Quantum Approximate Optimization Algorithms (QAOA)

The QAOA was gradually developed by [Farhi and Harrow \(2016\)](#) from an initial state Equation (2), which is the parent input authenticating the encoding process and is widely adopted by QAOAz and QWOA algorithms, and further forms the basis of our new QMOA scheme. The Hamiltonian C is defined as $U_C(\gamma) = \exp^{-i\gamma C}$ and encodes the values of the objective function $c(z)$ as its diagonal elements ([Slate et al., 2021](#)). The entries of the mixing operator U_{mix} vary depending on the relationship between solution states. This algorithm recommends a lower expected value, which corresponds to a higher probability measure in terms of the portfolio giving optimal or near-optimal future returns and risk.

Given any ordered random variable $X^{(i)}$, which is $S_{(i)}$ asset for our case, the mixing operator U_{mix} of the original QAOA scheme, the soft constraint $\zeta(z)$ (addition of a penalty function), and the initial state for the soft constraint method where $N^* = 2^{2N}$ ([Farhi and Harrow, 2016](#)) are defined in Equation (21), respectively.

$$U_{QAOA}(t) = \exp^{-it \sum_i X^{(i)}}, \zeta(z) = \varepsilon \left(A - \sum_{i=1}^N z_i \right)^2, \text{ and } |\psi_0\rangle = \frac{1}{\sqrt{2^{2N}}} \sum_{i=0}^{N^*-1} |i\rangle. \quad (21)$$

4.2. Quantum Alternating Operator Ansatz (QAOAz)

The QAOAz scheme is an extension of the QAOA that modifies the mixing operator to accommodate the optimization process to the subspace of valid solutions ([Slate et al., 2021](#)). Here, the mixing operator U_{QAOAz} is approximated from the “dual parity ring” Hamiltonian denoted by H_{ring} ([Hodson et al., 2019a](#)) (see Equation (22)).

$$H_{ring} = \sum_{a=1}^{2N} \left(X^{(a)} X^{(a+2)} + Y^{(a)} Y^{(a+2)} \right). \quad (22)$$

All the addition is processed modulo N . The Hamiltonian holds when 1) the hamming weight for both U and D qubits choices are independently preserved and 2) it meets the condition of constraint A (see Equation (20)). In this study ([Hadfield et al., 2019](#)), we devised a scheme that approximates the time evolution of the dual parity ring mixer by decomposing the mixer operator into three non-commuting unitaries.

The mixing operator U_{QAOAz} preserves both the upward and downward qubits independently by creating a structure called “parity bands,” described in [Hodson et al. \(2019b\)](#). For instance, if $N = 10$ and $A = 9$, then there are two possible valid portfolios, either 9 upward positions and 1 no position (H) or 10 upward positions and 1 downward position.

4.3. Quantum Walk Optimization Algorithm (QWOA)

The QWOA scheme is a generalization of the popular QAOA mixer as a continuous-time quantum walk over a subspace of valid solutions ([Marsh and Wang, 2020](#)). The advantages of the QWOA scheme are as follows:

- It is flexible in connecting only the solutions in the valid subspace,
- It reduces the search space by eliminating the degenerate portfolio states,
- It can eliminate the bias of one valid solution over another owing to mixing asymmetry. This helps in completing the global symmetry among other valid solutions.

QWOA implementation was conducted by designing an efficiently computable bijection function that maps the desired subspace of solutions into qubits set of the portfolio assets. This was inspired by the numerical results and performance of QWOA (see [Slate et al., 2021](#)).

4.4. Quantum Mix Optimization Algorithm (QMOA)

Each portfolio must be encoded using a quantum register, which is shown in the following table:

Table 1 summarizes the possible asset positions as a decision choice using qubits. For instance, “|01>” encoding implies that Choice U is strictly preferred to Choice D, whereas “|11>” implies that neither choices U nor D is selected (H position). QMOA advances the structure and builds the strength of three quantum combinatorial optimization algorithms (QAOA, QAOAz, and QWOA) (see **Table 2**).

First, A is the net sum of asset positions $z_i \in (1, -1, 0)$ in the trajectory. From **Table 1**, portfolio selection whose asset position is “|11>” is configured and equivalently interpreted as Choice H (degenerate position), whereas “|00>” is interpreted as no position (stationary state). For example, the yellow path in **Fig. 2** is a degenerate path whose assignment is “111111000000₂.” The QMOA search subspace is defined by the number of valid nondegenerate positions given the parameters (N, A) . This number defined in Equation (23) is equivalent to the number of lattice paths from the initial to final state (Slate et al., 2021).

$$\mathcal{M}(N, A) = \sum_{j=0}^N \binom{N}{j} \left(\frac{N-j}{\frac{1}{2}(N+A-j)} \right) \quad (23)$$

The second binomial coefficient in Equation (23) is set to zero if its bottom parameter is not an integer. In correspondence to the lattice in **Fig. 2**, every trajectory can be described using qubits by tracing the asset positions along the path. The assigned qubits set for the red path is “01101001011₂,” whereas the string of the green path is “10010110101₂.” To understand how the assignment is done in **Fig. 2**, the trace of the red path is $S_{13} \rightarrow S_9 \rightarrow S_{10} \rightarrow S_{11} \rightarrow S_7 \rightarrow S_3 \rightarrow S_4$. Each stage of the cycle is a decision choice where “U = 01₂,” “D = 10₂,” and “H = 11₂.” The last two digits of all nondegenerate strings are “11₂” which denotes the final and nondecision move.

Proposition 9. Let C_N be the number of cycles in the lattice. Then, we state the following:

1. The nondegenerate paths have $C_N/2$ decision moves.
2. There are $C_N - 1$ decision moves from the initial to final state.

For degenerate paths, the $C_N/2$ strings are “11₂” and the rest are “00₂” (stationary position).

4.4.1. QWOA vs. QMOA designs

The QMOA improves the efficiency of the QWOA scheme by contracting the valid subspace solution set S exponentially in time by half. This is verifiable by comparing the QWOA network in the study published by (Slate et al., 2021) (lattice labeled “**Fig. 1**”) and our new QMOA network in **Fig. 2**.

Table 2 presents comparisons of search space and degree of complexity for each algorithm based on their computation capability and lattice structure. The number $V(N, A)$ of portfolio encodings by Slate et al. (2021), defined by Equation (24), reveals that QWOA search space is smaller than that of QAOAz. The new QMOA scheme further reduces the search space.

$$V(N, A) = \sum_{j=0}^N 2^j \binom{N}{j} \left(\frac{N-j}{\frac{1}{2}(N+A-j)} \right) = \left(\frac{2N}{N+A} \right). \quad (24)$$

If the degeneracies are eliminated (ignored), then $A = 0$, which implies that out of $C_N - 1$ decision moves, half of it is upwards and the other half is downwards, with the final move being is “11”. Hence, $A = 0$ reduces the second binomial term in Equation (24) to N , resembling the theoretical results discovered in Equation (13). We then apply **Theorem 5** to Equation (24) when $A = 0$ to incorporate the possibility of an induced tadpole network.

Focusing on the p value in Equation (2), the quality of the solution keeps improving with an increase in p (Slate et al., 2021). To estimate the supremum of the p value in the QWOA model, we traditionally define the number of iterations in Equation (25), where k_c for $c \in [1, \dots, C_N]$ is the number of edges traversing cycle c .

Table 1

Qubit encoding of the possible asset positions (Slate et al., 2021).

“Choice U” qubit	“Choice D” qubit	z_i value	$ z\rangle$ encoding
0>	0>	0	00>
0>	1>	1	01> ^a
1>	0>	-1	10> ^b
1>	1>	0	11>

^a This is an upward move encoding on the lattice.

^b This is a downward move encoding on the lattice.

Table 2
Comparison of the search space and gate complexity.

	QAOA	QOAz	QWOA	QMOA (New)
Search Space	2^{2N}	$\binom{2N}{N+A}$	$\mathcal{M}(N, A)$	$\mathcal{M}(N, A)$
Gate Complexity	$\mathcal{O}(N)$	$\mathcal{O}(N)$	$\mathcal{O}(N^2)$	$\mathcal{O}(2N^2)$

$$T_{tt} = k_1 + \cdot k_2 \cdot \dots \cdot k_{C_N}. \quad (25)$$

Let $S(k_{C_N})$ be the set of all the possible product of k_c iterations derived from the lattice (see Equation (26)).

$$S(k_{C_N}) = \{\{k_1\}, \{k_1 \cdot k_2\}, \{k_1 \cdot k_2 \cdot k_3\}, \{k_1 \cdot k_2 \cdot k_3 \cdot k_4\}, \dots, \{T_{tt}\}\} \quad (26)$$

Therefore, the supremum of p belongs in $S(k_{C_N})$ with an upper bound of T_{tt} (see Equation 27).

Corollary 10. *The p value in QWOA falls in the range*

$$2 < p \leq T_{tt} \quad (27)$$

To calculate the number of QMOA iterations (p value), we consider the minimum number F of assets in the portfolio choice. As the number of iterations increases, the quality of the solution improves. For instance, if $F \leq C_N + 1$, QMOA produces the best results in only two iterations. Generally, the QMOA model produces superior results after the minimum g^{th} iteration such that

$$(g-1)(C_N-1) \leq F - \omega - 2 \leq g(C_N-1) \quad \forall g \geq 2. \quad (28)$$

Equation (28) provides a fundamental background in deriving and verifying QMOA iterations, p . Disregarding the degeneracies (H-moves), the upper bound of p for the QMOA iterations is derived from Theorem 5 by eliminating the ω nodes in the path graph. Hence, the possible number of iterations for the nonpath graph is defined by Equation (29).

$$I_p^* = \binom{2(n-1)}{(n-1)} = \frac{(2n-2)!}{((n-1)!)^2} \quad \text{such that } N - \omega = n^2. \quad (29)$$

Corollary 11. *The QMOA iterations, p , ranges between;*

$$2 \leq p \leq I_p^* \quad (30)$$

Proof. There is no rigorous proof of Corollary 11; however, this study attempts to illustrate it. Let I_F be the possible number of iterations involved in selecting $F \leq N$ number of assets to invest in a portfolio choice problem. Clearly, $2 \leq p \leq I_F$ and $I_F \leq I_N^*$. However, from the proof of Theorem 5, I_N^* is the minimum number of QMOA iterations, and I_p^* is the minimum number of QMOA iterations without the path graph in consideration. Hence, $I_F = I_p^*$, and therefore, $2 \leq p \leq I_p^*$.

To verify that the assets selected guarantee an optimal value in the portfolio choice problem, the objective function (Equation (20)) was also subjected to an ancillary constraint (Equation (31)). Let $S^*(k_{C_N})$ be a collection of the total number ($F \leq N$) of assets computationally selected for the portfolio. Furthermore, let $\sigma_{\min}^*$ be the mean value of covariances associated with the assets in set $S^*(k_{C_N})$ along the lattice path and σ_c denote the average weight of edges traversing the c^{th} cycle. Subsequently, the ancillary constraint to be met is defined by

$$\sigma_{\min}^* < \frac{1}{C_N} \sum_c^{C_N} \sigma_c \text{ or } C_N \sigma_{\min}^* < \sum_c^{C_N} \sigma_c. \quad (31)$$

We define a classical algorithm $\text{id}_{NA}(S_i)$, which indexes portfolio encodings of the state of assets. The following QMOA algorithm codes are generated to compute $\text{id}_{NA}(S_i)$ for any given portfolio asset S_i . For indexing, a simple recursion relation is used based on the counting function described in Equation (24). That is, by lattice analogy, the number of paths reaching (C_N, A) is equal to the sum of paths one step backwards from (C_N, A) coordinate. In other words, the one step backwards is either reverse-upward (A-1), reverse-downward (A+1), or reverse-horizontal (A). Hence, Equation (32) holds.

$$\mathcal{M}(C_N, A) = \mathcal{M}(C_N - 1, A + 1) + \mathcal{M}(C_N - 1, A - 1) + \mathcal{M}(C_N - 2, A). \quad (32)$$

Note that the right-hand side of 32 is $\mathcal{M}(C_N - 2, A)$ because the reverse-horizontal move skips a cycle in between and that this neutral move does not affect the net value of A .

4.4.2. Description of QMOA algorithm

The first line of coding initializes a random set $S^*(k)$ to hold the minimum covariances from each iteration process. Subsequently, we compute the least covariance in every cycle and append it to $S^*(k)$. Unless the number of assets is equal to the minimum required F , iteration continues until set $S^*(k)$ has sufficient elements that guarantee a better selection. This is attained by activating the CNOT gate (De Almeida et al., 2019), which ensures that the threshold of F minimum assets is met. Moreover, the initial condition is to satisfy $F - 1 - \omega = C_N + a$ for all $a \geq 0$, where $a - 1$ is the number of additional assets (excluding the ω assets) necessary to meet the minimum threshold F after the 1st iteration. After every iteration, the swap gate is activated, swapping assets within and across the cycles of the lattice. The swap is done to ensure there are no instances where cycles and their neighborhood have a lesser minimum weight in the subsequent iteration(s) as compared to the weight in the predecessor iteration. Note that each iteration involves C_N computations because there are C_N cycles in the network. The estimator operator $\mathcal{O}_\pi$ is then applied to recalculate the covariances of newly established edges between every pair of assets after the swap.

Subsequently, a string R concatenates U-moves and D-moves with 01_2 and 10_2 , respectively, as the algorithm traces the trajectory of the decision from one cycle to the next. These iterations occur until the set $S^*(k)$ contains sufficient elements required for the portfolio choice. Case 00_2 is considered a stationary position, and there is no decision made. Conversely, 11_2 implies that the decision neither favors U-move nor D-move; hence, the H-move is taken. Case 11_2 is rare and is carefully decided under the banner of an estimator function $\mathcal{O}_\pi$ (unbiased) applied to horizontally aligned assets in the lattice. Therefore, the wealth invested is distributed among $C_N/2 + 1$ assets as opposed to $C_N + 1$ assets for investment without a single H-move. Because the initial state, final state, and ω -assets are already known, the final two qubits 11_2 and ω are concatenated on the string R , forming string R^* that ensures the trajectory is complete. String R^* is then printed, and assets associated along its trace become the selected portfolio assets.

Algorithm 12.

QMOA Algorithm $id(N, A, S_0, S_N, C_N, k_c, \omega, F)$, $F - 1 - \omega = C_N + a \mid a \geq 0$

$S^*(k) = []$, recall ω Random set of min covariances
 for j in range $(1, k_c)$; 1st iteration
 $\sigma_j^1 = \min\{\text{covariance in cycle } c\}$
 append σ_j^1 to $(S^*(k))$
 $c = c + 1$
 for j in range $(1, k_c)$; 2nd iteration
 $\sigma_j^2 = \min\{\text{covariance in } c \text{ in the remaining set without } \sigma_j^1\}$
 append σ_j^2 to $(S^*(k))$, $c = c + 1$
 .
 .
 .
 for j in range $(1, k_c)$; t^{th} iteration
 $\sigma_j^t = \min\{\text{covariance in } c \text{ in the remaining set without } \sigma_j^1, \dots, \sigma_j^{t-1}\}$
 append σ_j^t to $(S^*(k))$ while $c = c + 1$
 return $S^*(k)$
 If $\text{len}(S^*(k)) \geq F - 1 - \omega$,
 return $S^*(k^*) \subseteq S^*(k)$ such that $\text{len}(S^*(k^*)) = F - 1 - \omega$
 $\iff \sum h \leq \sum m \quad \forall h \in S^*(k^*), \quad m \in S^*(k)$
 $\{S_i\} \longleftrightarrow^{y\text{-swap-}z} \{S_j\}$ where $y, z \in (1, C_N) \mid i \neq j, y \neq z$ Swap gate in
 multiaction
 $G \simeq \mathcal{O}_\pi\{(E_{p,q})\}$ Restructure G & apply $\mathcal{O}_\pi$ to edges $E_{p,q}$ for some p, q
 $G^* \leftarrow G \leftrightarrow w$ Attach T_ω
 trace from state S_0 to S_N via a and ω $a = \text{auxiliary assets}$
 Case 00_2 : $\emptyset \leftarrow \mathcal{M}(1, A = 0, S_0)$ & 00_2 $\triangleright$ Null-move
 Case 01_2 : $R \leftarrow \mathcal{M}(C_N, 0) + id(C_N - 1, -1, S_i)$ & 01_2 $\triangleright$ U-move
 Case 10_2 : $R \leftarrow \mathcal{M}(C_N, 0) + id(C_N - 1, 1, S_j)$ & 10_2 $\triangleright$ D-move
 Case 11_2 : $R \leftarrow \mathcal{M}(C_N, 0) + id(C_N - 2, 0, S_k)$ & 11_2 H-move
 Repeat Cases $00_1, 01_2, 10_2$, & 11_2 until $S_i = S_j = S_k = S_0$
 $R = S^*(k^*)$ where $\text{len}(S^*(k^*)) = F - 1 - \omega$
 $R^* \leftarrow R \lll 11 \oplus \omega$ Last two bits of R & concatenation of ω
 print "Assets along path $R^* = S(R)_F$ "

Key:

- Recall k_c is the number of edges traversing the c^{th} cycle, $c \in [1, C_N]$.

- $S^*(k^*)$ is the set containing the minimum number of assets in the portfolio required, that is, the minimum threshold.
- Case 11₂ skips 1 cycle per every move, thus taking $C_N/2$ moves to trace the path. Hence, the other half of the string path is concatenated with 00₂.

5. Conclusion

This study discusses the comparison of quantum combinatorial optimization algorithms for portfolio selection. These algorithms include the QAOA, QAOAz, QWOA, and our newly developed scheme QMOA. QMOA focuses on the portfolio selection process of discrete nature subject to a complex set of discrete constraints.

QMOA formulation starts by constructing a network where the nodes are assets and the edges are the corresponding covariances. Emphasis is placed on minimizing risk by selecting relatively small covariances. The investor first identifies two assets from the pool called the initial and final states. If the investor prefers over two assets (say ω more) to be included in the portfolio, then ω assets form a path graph of an induced tadpole network. The network is then formulated, fixing the other assets in between the initial and the final states. Subsequently, the network is partitioned into portions called “cycles,” and the minimum weight of edges traversing each cycle is computed (see Fig. 2). The first round of selection is attained by the 1st iteration. The algorithm traces the assets along the trajectory and displays them as the set to be considered in the portfolio.

However, if the 1st iteration does not provide sufficient assets for the portfolio, the 2nd process is iterated, and the algorithm determines the second minimum weight in every cycle. The “swap gate” is then activated to swap assets whose second minimum weights are less than the first minimum weights in other cycles. This occurs even in the subsequent iterations to strictly ensure the trajectory of minima is attained in the selection process. This demonstrates that the 1st iteration is not necessarily safer and does not fully guarantee the best selection. For $a = \omega = 0$, there are $C_N + 1$ assets, and the 1st iteration guarantees sufficient assets in $S^*(k)$. However, a second iteration provides the best solution based on Equation (28). Therefore, we recommend running $p = g + r$ (for some $r \geq 0$) iterations such that Equation (28) is satisfied.

From the theoretical analysis, QMOA results highlight key advantages over QAOA, QAOAz, and QWOA. First, the search space of QMOA is twice smaller compared to that of the QWOA, referring to the lattices of the two models. Second, the approach of QMOA partitions the solution space into subspaces called cycles, which effectively contributes to computational speed exponentially. Finally, compared to the other algorithms, QMOA has a relatively smaller range of p value, requiring only two iterations if $F \leq C_N + 1$ (see Corollaries 11 and 10, respectively).

QMOA provides no numerical results; hence, it would be interesting to investigate the performance of its experimental results compared to that of other schemes. A numerical analysis was not conducted owing to the inaccessibility of a quantum or hybrid (quantum-classical) laboratory to run the QMOA algorithm. However, QMOA algorithm codes are available for implementation. In addition, the unbiased estimator $\mathcal{O}_\pi$ is well-defined statistically. We recommend similar software used in QAOA, QAOAz, and QWOA algorithms. The software package *QuOPMPI* (Matwiejew and Wang, 2021; Slate et al., 2021) was used for QWOA and will probably work well for QMOA. The parameters $\vec{\gamma}$ and $\vec{t}$ can be estimated by Head and Zerner (1985) and datasets can be obtained from any financial institution, such as Yahoo Finance, using a Python-based API (see (Han and Neumann, 2006)). A brief description of QMOA algorithm codes is provided to aid the configuration, installation, and implementation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics Statement

Not applicable because this work does not involve the use of animal or human subjects.

References

- Basavanagoud, B., Desai, V. R., & Patil, S. (2017). () connectivity index of graphs. *Applied Mathematics and Nonlinear Sciences*, 2(1), 21–30.
- Cook, J., Eidenbenz, S., & Bärttschi, A. (2020). The quantum alternating operator ansatz on maximum k-vertex cover. In *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)* (pp. 83–92). IEEE.
- De Almeida, A. A., Dueck, G. W., & Da Silva, A. C. (2019). Cnot gate optimizations via qubit permutations for ibm's quantum architectures. *J. Low Power Electron.*, 15(2), 182–192.
- Farhi, E., & Harrow, A. W. (2016). *Quantum Supremacy through the Quantum Approximate Optimization Algorithm*. arXiv preprint arXiv:1602.07674.
- Hadfield, S., Wang, Z., O'gorman, B., Rieffel, E. G., Venturelli, D., & Biswas, R. (2019). From the quantum approximate optimization algorithm to a quantum alternating operator ansatz. *Algorithms*, 12(2), 34.
- Han, K.-H., & Kim, J.-H. (2000). Genetic quantum algorithm and its application to combinatorial optimization problem. In *Proceedings of the 2000 Congress on Evolutionary Computation. CEC00 (Cat. No. 00TH8512)* (vol. 2, pp. 1354–1360). IEEE.
- Han, L., & Neumann, M. (2006). Effect of dimensionality on the Nelder–Mead simplex method. *Optim. Methods Software*, 21(1), 1–16.
- Head, J. D., & Zerner, M. C. (1985). A broyden–fletcher–goldfarb–shanno optimization procedure for molecular geometries. *Chem. Phys. Lett.*, 122(3), 264–270.
- Hodson, M., Ruck, B., Ong, H., Garvin, D., & Dulman, S. (2019a). *Portfolio Rebalancing Experiments Using the Quantum Alternating Operator Ansatz*. arXiv preprint arXiv:1911.05296.

- Hodson, M., Ruck, B., Ong, H., Garvin, D., & Dulman, S. (2019b). *Portfolio Rebalancing Experiments Using the Quantum Alternating Operator Ansatz*. arXiv preprint arXiv:1911.05296.
- Hurkens, C. A., & Woeginger, G. J. (2004). On the nearest neighbor rule for the traveling salesman problem. *Oper. Res. Lett.*, 32(1), 1–4.
- Kim, W. C., Kim, J., & Fabozzi, F. (2015). *Mean-Variance Portfolio Selection* (vol. 11, pp. 6–21).
- Lawler, E. L. (1985). *The Traveling Salesman Problem: a Guided Tour of Combinatorial Optimization*. Wiley-Interscience Series in Discrete Mathematics.
- Markowitz, H. (1952). Portfolio selection. *J. Finance*, 7(3), 77.
- Marsh, S., & Wang, J. B. (2020). Combinatorial optimization via highly efficient quantum walks. *Physical Review Research*, 2(2), Article 023302.
- Marzec, M. (2014). *Portfolio Optimization: Applications in Quantum Computing* (p. 2).
- Matwiejew, E., & Wang, J. (2021). Qsw-mpi: a framework for parallel simulation of quantum stochastic walks. *Comput. Phys. Commun.*, 260, Article 107724.
- Milne, Andrew, & Maxwell Rounds, P. G. (2017). *Quantum Computational Finance: Monte Carlo Pricing of Financial Derivatives*. White Paper 1Qbit.
- Mugel, S., Kuchkovsky, C., Sanchez, E., Fernandez-Lorenzo, S., Luis-Hita, J., Lizaso, E., & Orus, R. (2020). *Dynamic Portfolio Optimization with Real Datasets Using Quantum Processors and Quantum-Inspired Tensor Networks* (p. 6).
- orus, R., Mugel, S., & Lizaso, E. (2019). Quantum computing for finance: overview and prospects. *Rev. Phys.*, 4(2), Article 100028.
- Rebentrost, P., Gupta, B., & Bromley, T. R. (2018). Quantum computational finance: Monte Carlo pricing of financial derivatives. *Phys. Rev.* (98) p. 022321.
- Renumadhavi, C., Srivastava, A., Singh, A. K., & Mittal, A. (2021). *Quantum Finance an Overview*. EasyChair Preprint No.6071.
- Slate, N., Matwiejew, E., Marsh, S., & Wang, J. (2021). Quantum walk-based portfolio optimisation. *Quantum*, 5, 513.
- Weisstein, E. W. (2007). *Tadpole Graph*. <https://mathworld.wolfram.com/>.